Corrected and conduct of chapter 3

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. $(3; -4)$. *Action:* Subtract arrival less departure on two drawings.
2. $\sqrt{3^2 + (-4)^2} = 5$. *Action:* Review Pythagoras and distinguish norm and coordinates.
3. $\overrightarrow{AC}$. *Action:* Trace the compound displacement.
4. $a^2 + 2ab + b^2$. *Action:* Identify the cross-word.
5. $\frac{3}{5}$. *Action:* Name the sides in relation to the angle.
6. $(1; 2)$. *Action:* Measure the two coordinates separately.

Research activity

Take points on the AB diameter circle; demonstration by midpoint I: product $MI^2 - 9$.
Extremities are included because one of the vectors is zero.

Immediate applications

1. 0 then -10. Cosine is equal to 0 and -1.
2. $3t - 12 = 0$, therefore $t = 4$.
3. The square is worth $4 - 2 + 9 = 11$; the norm is $\sqrt{11}$.
4. $32 = 16 + 16 - 32\cos A$, therefore $\cos A = 0$ and $A = 90°$.
5. $MI^2 - 9 = 7$: centre circle (3; 0) and radius 4.

Exercises: approach and results

Fiscal year 3.1 · Direct calculations. Output $4 - 6 = -2$; norms $\sqrt{5}$ and 5; sum (5; 1) squared 26.
Control: $5 + 2(-2) + 25 = 26$.

Exercise 3.2 · A right triangle?. $\overrightarrow{AB} = (3; 1)$, $\overrightarrow{AC} = (-1; 3)$; product 0. $AB = AC = \sqrt{10}$, area $\frac{AB \times AC}{2}$ = 5 units of area.

Exercise 3.3 · Length without reference. $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{AD}$ and $\overrightarrow{BD} = \overrightarrow{AD} - \overrightarrow{AB}$. Output $\frac{4 \times 3}{2} = 6$; $AC^2 = 16 + 9 + 12 = 37$ and $BD^2 = 16 + 9 - 12 = 13$.

Exercise 3.4 · Find an angle. $\cos\theta = -\frac{6}{12} = -\frac{1}{2}$; in [0; 180 °], the angle is 120°. The field ensures uniqueness.

Exercise 3.5 · A variable location. $MI^2 = k + \frac{AB^2}{4}$. If the second member is negative: together empty ; zero : the only point I ; positive : radius circle its square root. The discussion is about a distance to square.

Exercise 3.6 · Force work. $W = \frac{50 \times 4 \times 1}{2} = 100$ J. À $120°$, $W = -100$ J. The sign Distinguishes aid for the displacement of opposition; it does not describe a negative distance.

Exercise 3.7 · Contrapositive. Direct "orthogonal" ownership implies equality of squares The contrast is exactly stated. You can also write 2 $\vec{u} \cdot \vec{v} \neq 0$.

Fiscal year 3.8 · Deferred review. Al-Kashi: $\frac{9 + 16 - 25}{2}$ = 0. Pythagoras Converse: Triangle right-angled at A, therefore zero product. Name the assumptions of each method.

CQFD, exercise 9 p. 250, questions a and b. a : $1 \times 3 + 2 \times (-1) = 1$. b : $(-1) \times 12 + 3 \times 7 = 9$. The scalar product is a number, not a vector.

CQFD, Exercise 11 at 250, Questions 1, 2 and 3. $\overrightarrow{AB} = (4; 6)$, $\overrightarrow{AC} = (-3; 4)$, $I(-1; -2)$ and $\overrightarrow{CI} = (5; -(1)$ Where $\overrightarrow{AB} \cdot \overrightarrow{AC} = 12$, $\overrightarrow{AB} \cdot \overrightarrow{CI}$ = 14. Chasles gives $\overrightarrow{AB} \cdot \overrightarrow{AI}$ = 12 + 14 = 26. controls: $\overrightarrow{AI}$ = (2; 3) gives 8 + 18 = 26; or $\overrightarrow{AI} = \frac{1}{2}\overrightarrow{AB}$ gives $\frac{1}{2}(16 + 36) = 26$.

MCQ: answers and reasons

MCQ 1.

1. A. $6 - 4 = 2$.

2. C. Root of 25.

3. D. Scalar criterion.

4. B. $\frac{2 \times 3}{2} = 3$.

5. C. Developing the scalar square.

MCQ 2.

1. D. Negative Cosine.

2. A. Projection coordinates.

3. C. $MI^2 = \frac{AB^2}{4}$.

4. B. Bilinearity: 2($-1)3$.

5. D. $BC^2 = 8$.

True or false

1. False: $u \cdot (-u) = -\|u\|^2 < 0$ for $u \neq 0$.

2. True: its product with any vector is 0.

3. False: (1; 0) and (0; 1) are non-zero and orthogonal.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.